How Much of a Time Series Survives Its Image? A Cross-Domain Comparative Analysis of the Reconstruction Capacity of Image-Based Representations

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Abstract

Image-based representations of time series are used as classifier inputs and, increasingly, as the space in which signals are generated or restored, which requires mapping an image back to a series. We compare how much of a series each representation retains. We study the most widely used ones, the recurrence plot (RP) in its binary, unsigned and signed forms, the Gramian angular summation and difference fields (GASF, GADF) and the Markov transition field (MTF), together with the recently proposed XIRP, whose diagonal stores the series. Each is decoded from its 8-bit image alone, with the published inverse where one exists and an operational decoder where none does. On seven datasets from five domains, we measure temporal and spectral fidelity, robustness, cost, the conditions for exact reconstruction, and whether classifiers trained on original signals keep their accuracy on reconstructions. The signed RP with a geometric decoder, GASF, GADF and XIRP are exact in floating point and keep classifier accuracy, losing on average less than 0.1 percentage points; binary and unsigned RP and MTF lose most of the

waveform. The sign recovers the RP's orientation, and the conditioning of its states, which degrades as the sampling rate grows, governs its error.
Keywords: Time-series imaging, Reconstruction, Recurrence plots, Gramian angular fields, Markov transition fields, Invertibility, Comparative analysis

1. Introduction

Turning a time series into an image has become a common way of giving image-domain models access to temporal data. Recurrence plots (RP), Gramian angular fields (GAF) and Markov transition fields (MTF) place a relation between every pair of time instants in a matrix, and convolutional and other image models then learn from that matrix [1, 2, 5, 6]. The same representations are now also used in the opposite direction: a generative or restoration model works on images, and its output has to be turned back into a series [3, 4].

That second use depends on a property the first never needed. A classifier only has to separate classes, and it can do so from an image that has discarded amplitude, phase or temporal order. A model that produces images is useful only if those images can be decoded into the signal they stand for. Whether, and how well, each representation allows this has not been compared: reconstruction is usually studied for one representation at a time, on one kind of signal, and is often judged only through the accuracy of a downstream classifier.

This paper is a comparative analysis of the reconstruction capacity of image-based time-series representations. We take the most widely used ones, the recurrence plot, the Gramian angular summation and difference fields

(GASF, GADF) and the Markov transition field, and add the extended intertemporal return plot (XIRP) of Hellermann and Lessmann [4], a recent representation proposed for image-based generation that stores the series itself on its diagonal and is therefore recoverable by construction. For the recurrence plot we separate the encoding from the decoder: we compare the binary RP of the classical literature, the continuous unsigned RP and the signed RP of Lu and Tong [11], decoded both with the classical reconstruction of Hirata et al. [12] and with a geometric decoder.

The setting is controlled. Every image is the deterministic output of a known encoder, stored with 8 bits per pixel, and every decoder works from the stored image alone. Where a representation has a published inverse we use it; where it has none that can be applied to a stored image, as for the signed RP and GADF, we provide one, since otherwise the comparison could not be made. Decoding the floating-point matrix as well as the 8-bit image separates what the representation loses from what storage loses. Figure 1 summarises this round trip.

The analysis answers five questions:

- *RQ1.* How much of the signal can be recovered from each representation, in time and in frequency, and how much of the loss is due to 8-bit storage?
- *RQ2.* For recurrence plots, what do the encoding and the decoder each contribute?
- *RQ3.* Under which conditions is reconstruction exact, and how does it vary with the domain, the sampling rate and the window length?
- *RQ4.* Do reconstructions preserve what classifiers trained on the original

signals need?

- *RQ5.* What does each reconstruction cost in time and memory?

The contributions are:

- A comparative analysis of the reconstruction capacity of five image representations, eight encoding–decoder combinations in total, on seven public datasets from five domains (human activity, environmental audio, electrocardiography, geomagnetism and seismology) and three window lengths, under one protocol of fidelity, cost and classification utility.
- A separation of the two ingredients of RP reconstruction, encoding and decoder, which shows what the sign of the signed RP adds and what the decoder adds.
- A statement of what each image determines and of the conditions under which it is exactly, approximately or only partially invertible, with the conditioning of the recurrence-plot states identified as the quantity that governs reconstruction from the signed RP after storage.
- Operational, metadata-free decoders for the two representations that lacked one, the signed RP and GADF, and the code and derived data needed to reproduce every result.

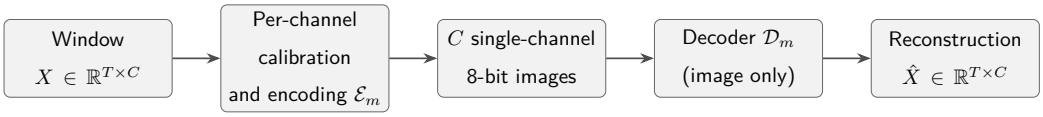


Figure 1: Controlled reconstruction. Each channel is calibrated, encoded and stored as an 8-bit image, then decoded from the image alone; the shared per-channel calibration returns it to physical units. Decoding the floating-point operator instead of the image isolates the algorithmic error from the storage error.

2. Related work

2.1. Image representations of time series

Recurrence plots were introduced to visualise when a dynamical system revisits a region of its state space [5] and have since been developed into a family of quantitative tools [6]. In their conventional form they are binary: a pixel marks whether two embedded states lie within a threshold of each other. Continuous variants keep the distance itself, and the signed variant of Lu and Tong also keeps whether the displacement between two states follows a reference direction, which they used for activity recognition from a single accelerometer [11]. Wang and Oates proposed the Gramian angular fields, which map each sample to an angle and store the cosine of their sum or the sine of their difference, and the Markov transition field, which stores transition probabilities between quantile states, as image inputs to convolutional networks [1, 2]. These representations have been applied to human activity recognition [7], anomaly detection [8] and imputation [9]. Hellermann and Lessmann recently introduced XIRP for time-series generation with image-based generative adversarial networks; it stores log-returns between instants off the diagonal and the series on the diagonal, so that generated

images can be inverted [4].

2.2. Recovering a series from its image

Inversion has been studied separately for each representation. For binary recurrence plots, Hirata et al. build a graph from the recurrence neighbourhoods, approximate the distances between states by shortest paths and recover the state geometry by multidimensional scaling [12, 26]. For unthresholded recurrence plots of delay embeddings, Sipers et al. characterise when the signal is identifiable up to its mean and sign and reconstruct it by optimisation [13, 14]. For GASF, Wang and Oates read the series from the diagonal, and restore damaged images with a denoising autoencoder before doing so [1]; Tao et al. follow a similar route for RP and GAF images [9]. For GADF no decoder that resolves its branch ambiguity and its lost global offset has been specified, and for MTF even approximate recovery has been reported to be difficult [1]. XIRP is inverted from its diagonal or by averaging the proposals of all its columns [4].

2.3. Gap

Each of these works evaluates one representation, usually on one kind of signal, and several evaluate reconstruction only through classification. No study has compared the reconstruction capacity of the representations under a common protocol, separated what a representation loses from what image storage loses, or distinguished, for recurrence plots, the contribution of the encoding from that of the decoder.

3. Mathematical formulation

3.1. Encoding, storage and decoding

Let $x = (x_1, \ldots, x_T)$ be one channel of a window. Every method m applies a calibration to a method-specific normalised domain, an encoder $\mathcal{E}_m$ producing a matrix $M^{(m)}$, a fixed 8-bit storage map $\mathcal{Q}_m$ and a decoder $\mathcal{D}_m$:

$$\hat{x} = \mathcal{N}^{-1}\Big(\mathcal{D}_m(\mathcal{Q}_m(\mathcal{E}_m(\mathcal{N}(x)))); \, x_{\max}, x_{\min}\Big).$$

The calibration $(x_{\max}, x_{\min})$ is shared by all methods and is the only per-channel information used to return to physical units. We call a representation *exactly invertible* when $\mathcal{D}_m \circ \mathcal{E}_m$ is the identity on its normalised domain, so that all error in $\hat{x}$ is due to $\mathcal{Q}_m$; *approximately invertible* when $\mathcal{D}_m \circ \mathcal{E}_m$ is not the identity but recovers the series up to a bounded algorithmic error; and *lossy* when $\mathcal{E}_m$ is many-to-one.

3.2. Recurrence plots

With the adjacent states $s_i = (x_i, x_{i+1})$, $i = 1, \ldots, T-1$, and $v = (1,1)/\sqrt{2}$, the continuous signed RP [11] is

$$R_{ij} = \sigma_{ij}\|s_i - s_j\|_2, \qquad \sigma_{ij} = \begin{cases} +1, & \|s_i - s_j\| = 0 \text{ or } \hat{u}_{ij} \cdot v \geq \cos(\pi/4), \\ -1, & \text{otherwise}, \end{cases} \tag{1}$$

with $\hat{u}_{ij} = (s_i - s_j)/\|s_i - s_j\|$. The unsigned RP is $|R|$, and the conventional binary RP is $B_{ij} = \mathbb{1}[\|s_i - s_j\| < \varepsilon]$ [5, 6]. A pair carries directional information only when its displacement lies within $\pi/4$ of $\pm v$: then $\sigma_{ij} = -\sigma_{ji}$, while displacements near the anti-diagonal receive -1 in both orders.

Proposition 1 (Signed RP, floating point)**.** *Let $S \in \mathbb{R}^{(T-1)\times 2}$ be the matrix of states and assume that the centred states span $\mathbb{R}^2$, i.e. $\sigma_2(S - \bar{S}) > 0$. Then classical MDS applied to $|R|$ recovers S up to a translation and an orthogonal transformation $Q \in O(2)$. If, in addition, the adjacent-state overlap $b_i = a_{i+1}$ is satisfied by a unique orientation up to the sign selection of Section 4.2, the decoder returns x exactly on the normalised domain.*

Proof sketch. The first part is the classical result that an exact Euclidean distance matrix of points with affine rank two is reproduced up to isometry by the two leading eigenvectors of the doubly centred squared-distance matrix [15]. For the second, the overlap residual is linear in $(\cos\theta, \sin\theta)$ for each reflection parity (Section 4.2), so the exact orientation attains zero residual. [pending: complete proof and the uniqueness condition, issue #121]

Remark 1 (8-bit storage)**.** *Storage perturbs $|R|$ by at most half a quantisation step relative to its range. The error of the recovered two-dimensional subspace grows as the gap between the second and third eigenvalues of the centred Gram matrix shrinks, so the reconstruction degrades as σ_2/σ_1 of the centred states tends to zero, i.e. as the states become collinear [27]. Empirically, no series with $\sigma_2/\sigma_1 \geq 0.1$ exceeds a normalised error of* 0.01 *(Section 6.3).*

3.3. Gramian angular fields

After scaling to the method-specific interval and $\phi_i = \arccos \tilde{x}_i$, the summation and difference fields are $G^{\mathrm{S}}_{ij} = \cos(\phi_i + \phi_j)$ and $G^{\mathrm{D}}_{ij} = \sin(\phi_i - \phi_j)$ [1]. On $[0, 1]$, $G^{\mathrm{S}}_{ii} = \cos(2\phi_i)$ determines $\tilde{x}_i$ uniquely, so GASF is exactly invertible. GADF has $G^{\mathrm{D}}_{ii} = 0$; on the symmetric interval $[-\alpha, \alpha]$ the inverse sine returns $\phi_i - \phi_j$ without branch ambiguity if and only if $\alpha < 1/\sqrt{2}$, and the remaining

global offset is fixed by an image-derived boundary anchor (Section 4.4). Under these hypotheses GADF is exactly invertible.

3.4. Markov transition fields and XIRP

MTF assigns every sample to one of Q quantile states and stores the transition probability between the states occupied at two positions [1]. Samples in the same state are indistinguishable, so the operator is many-to-one and only a representative trajectory can be recovered. XIRP [4] requires a strictly positive series and stores $R_{ii} = x_i$ and $R_{ij} = \log(x_i/x_j)$; it is exactly invertible from its diagonal, and inversion by mean averages the T column proposals $x_j \exp(R_{ij})$.

3.5. What each image determines

Table 1 summarises, for each representation, what the image determines, what the decoder needs, and its class of invertibility.

4. Reconstruction methods

Table 2 lists the decoder used for each representation in the comparison. Published inverses are used as described by their authors, with only the adaptations needed to decode a stored 8-bit image; the signed-RP and GADF decoders are ours, because neither representation had a decoder that could be applied to a stored image.

4.1. Storage protocol

Each matrix is written to $[0, 255]$ by a fixed or per-window affine map. GASF and GADF are read back with the fixed operator range $[-1, 1]$, MTF

Table 1: Information determined by each image and the resulting class of invertibility. Every decoder uses the image alone on the normalised domain; returning to physical units uses the shared calibration $(x_{\max}, x_{\min})$.

Representation	Image determines	Invertibility
Signed RP	State geometry and directional sign	Exact if $\sigma_2/\sigma_1 > 0$ (Prop. 1)
Unsigned RP	State geometry only	Exact up to reflection
Binary RP	Recurrence neighbourhoods	Approximate (graph geodesics)
GASF	Angles on one branch	Exact on $[0, 1]$
GADF	Angular differences	Exact if $\alpha < 1/\sqrt{2}$
MTF	Quantile states and transitions	Lossy
XIRP	Series on the diagonal	Exact

with $[0, 1]$, and XIRP with the fixed ranges of its diagonal and off-diagonal parts after the series is mapped to $[1, 2]$. The signed RP keeps a per-window map; its zero-valued diagonal locates $R = 0$, so subtracting the median diagonal pixel recovers the signed matrix up to a positive scale that MDS and the final range map absorb. No decoder uses per-window metadata.

4.2. Signed-RP decoder

1. Recover $R^{\text{obs}} = \text{PNG} - \text{median}_i \, \text{PNG}_{ii}$ and the symmetric distances $D_{ij} = \frac{1}{2}(|R^{\text{obs}}_{ij}| + |R^{\text{obs}}_{ji}|)$.
2. Classical MDS [15]: with $J = I - \mathbf{1}\mathbf{1}^\top/(T-1)$, the two leading eigenpairs of $-\frac{1}{2} J D^{\circ 2} J$ give $Y = V_2 \Lambda_2^{1/2}$.

Table 2: Encoding and decoder of every compared combination.

Encoding	Decoder	Origin of the decoder
Binary RP	Graph geodesics and MDS	Hirata et al. [12]
Signed RP	Hirata, sign used for the reflection	Hirata et al., adapted
Unsigned RP	Two-dimensional MDS and overlap	This work
Signed RP	Two-dimensional MDS, overlap and sign	This work
GASF	Diagonal inverse	Wang and Oates [1]
GADF	Restricted branch and boundary anchor	This work
MTF	Fiedler-ordered representative	This work (lossy)
XIRP	Inversion by mean	Hellermann and Lessmann [4]

3. Orientation in closed form. Every $Q \in O(2)$ is $\mathrm{rot}(\theta)\,\mathrm{diag}(1,\eta)$ with $\eta \in \{\pm 1\}$. Writing $Z = YQ$ and $(a_i, b_i) = (Z_{i1}, Z_{i2})$, the overlap $b_i = a_{i+1}$ must hold up to a common translation δ. For fixed η the residual $b_i - a_{i+1} = A_i \cos\theta + B_i \sin\theta$ with $A_i = \eta Y_{i2} - Y_{i+1,1}$ and $B_i = -\eta Y_{i1} - Y_{i+1,2}$; profiling out δ centres A and B, and the objective becomes the Rayleigh quotient $u^\top H_\eta u$ with $H_\eta = \tilde{C}^\top \tilde{C}$, $\tilde{C} = [\tilde{A}\ \tilde{B}]$ and $u = (\cos\theta, \sin\theta)$. Its minimiser is the eigenvector of the smaller eigenvalue of the 2×2 matrix H_η, so the two parities give four exact candidates.
4. Sign selection. Among the four candidates, keep the one whose regen-

erated signed RP is closest to R^{obs} in Frobenius norm after both are normalised to unit norm. The orientation is chosen from the image alone; the original series is never consulted.

5. Samples: $\hat{x}_1 = a_1$, $\hat{x}_i = \frac{1}{2}(b_{i-1} + a_i)$ for $2 \le i \le T-1$, $\hat{x}_T = b_{T-1}$, and the minimum and maximum are mapped to the normalised range.

The orientation costs $O(T)$ and the decoder is dominated by the eigendecomposition, $O(T^3)$ in a dense implementation. On the unsigned RP the same decoder runs without step 4, so the reflection remains undetermined.

4.3. Classical reconstruction of Hirata et al.

As the reference for the conventional RP we implement the reconstruction of Hirata et al. [12, 26]: from the binary RP, an edge joins $i \neq j$ when $B_{ij} = 1$, with local length $1 - |G_i \cap G_j|/|G_i \cup G_j|$, $G_i = \{k : B_{ik} = 1\}$; all-pairs shortest paths give a distance matrix, and classical MDS gives the states. The threshold ε is not prescribed by the method; we set it to a recurrence rate of 0.75. Turning the states into T samples is our post-processing: the leading MDS coordinate, the mean appended to reach T samples, and the range map. A binary RP carries no sign, so the reflection is left as returned. When Hirata's method is applied to the signed RP, its graph sees only the thresholded magnitude, so the sign is used, as in our decoder, only to choose between the series and its reflection.

4.4. GASF, GADF, MTF and XIRP decoders

GASF uses the diagonal inverse $\hat{\hat{x}}_i = \cos(\frac{1}{2}\arccos G^{\mathrm{S}}_{ii})$ [1]. For GADF on $[-\alpha, \alpha]$ with $\alpha = 0.7$, the row means $\mu_i = \frac{1}{T}\sum_j G^{\mathrm{D}}_{ij}$ are strictly increasing in ϕ_i, so $e = \arg\max_i |\mu_i|$ identifies a boundary sample and the sign of μ_e fixes

Table 3: Datasets. Windows of 128 samples available after segmentation; the group is the unit that never appears on both sides of a split.

Dataset	f_s (Hz)	Channels	Task	Windows	Group
WISDM	20	3	5 activities	8 172	subject (51)
PAMAP2	100	9	12 activities	15 085	subject (9)
ESC-50	8 000	1	50 sound classes	624 000	source recording
PTB-XL	100	12	5 classes, multilabel	21 000	patient
Swarm MAG	1	3	none	62 757	–
STEAD	100	3	earthquake vs noise	92 000	event / station
EMG gestures	100	8	6 gestures	880	subject (36)

which boundary; the angles follow from $\hat{\phi}_j^{(p)} = \phi_e - \arcsin G^{\mathrm{D}}_{ep} - \arcsin G^{\mathrm{D}}_{pj}$, averaged over the five pivots with the smallest $|\mu_p|$. MTF ($Q = 4$) is decoded to a representative trajectory by clustering the rows into Q states, ordering them by the Fiedler vector of the symmetrised transition matrix [16] and assigning each state its bin midpoint. XIRP uses inversion by mean [4].

5. Experimental protocol

5.1. Datasets

Table 3 lists the seven public datasets. They span human activity (WISDM [17], PAMAP2 [28]), environmental audio (ESC-50 [29]), electrocardiography (PTB-XL [30]), geomagnetism (Swarm [31]), seismology (STEAD [32]) and surface electromyography (EMG [33]), with sampling rates from 1 Hz to 8 kHz and one to twelve channels.

5.2. Windows and splits

Every channel is encoded, stored and decoded independently, as its own single-channel 8-bit image with its own calibration, whatever the number of channels: one image for ESC-50, three for WISDM, Swarm and STEAD, and eight, nine and twelve for EMG, PAMAP2 and PTB-XL. Channels are neither packed into colour planes nor combined, so results are comparable across datasets and cost grows linearly with the number of channels; representations that couple channels are outside the scope of this study. Each recording is cut into non-overlapping windows of $T \in \{64, 128, 256\}$ samples that never cross a label change or a time gap. For fidelity, 500 windows per dataset and length are drawn with a fixed seed. For classification, up to 6 000 windows per dataset and length are drawn, and three grouped folds keep every subject, patient, event, station or source recording on one side; folds are stratified by class for single-label data. Blocks with fewer than 200 windows (EMG at $T = 256$, where a gesture lasts about three seconds) are not reported. ESC-50 at 128 samples spans 16 ms of sound; audio is therefore also evaluated on the official ESC-10 subset at 2, 4 and 8 kHz with windows of 512 and 1 024 samples, silent windows removed and accuracy also scored per clip.

5.3. Fidelity, cost and utility

Fidelity is measured per channel series with range-normalised errors, writing $r = \max_i x_i - \min_i x_i$:

$$\mathrm{NMAE} = \frac{1}{rT} \sum_i |x_i - \hat{x}_i|, \qquad \mathrm{NRMSE} = \frac{1}{r} \sqrt{\tfrac{1}{T} \sum_i (x_i - \hat{x}_i)^2},$$

together with Pearson correlation, the range-normalised error of first differences, the L_1 distance between unit-mass Welch spectra [18], the dominant-

frequency error and the energy error. Constant series are counted and excluded. Each decoder is run on the floating-point operator and on the 8-bit image, which separates algorithmic from storage error. Cost is the single-thread time to encode and decode and the peak traced memory, measured for T from 64 to 1 024.

Robustness follows a local-stability protocol: zero-mean Gaussian noise of $\sigma \in \{2, 4, 8, 16\}$ intensity levels is added to every pixel of the stored image, which is clipped to $[0, 255]$, rounded and decoded, with two independent replicates per level on 200 windows per dataset and length. The sampling-rate study reads the same 3 000 PTB-XL records at 500 Hz and resamples them with an anti-aliased polyphase filter to 250, 100 and 50 Hz.

Utility follows the train-on-original protocol: KNN, SVM, random forest, LSTM, GRU and a one-dimensional CNN are trained on original windows and evaluated on the original and on every reconstruction of the same held-out windows.

5.4. Statistical protocol

All comparisons are paired over the same series. Differences between decoders are summarised by medians with 95% bootstrap intervals that resample whole groups [21, 22], tested with the Wilcoxon signed-rank test [19] and Holm's correction within each family [25], and averaged across datasets with equal weight. Classification equivalence requires the 95% group-bootstrap interval of the accuracy difference to lie within ± 0.01 [23].

6. Results

6.1. Fidelity across representations and domains

Table 4 gives the 8-bit reconstruction error of every combination at $T = 128$; the other lengths behave alike and are in the Supplementary Material. Over the 20 reported dataset–length blocks, 54 336 channel series were reconstructed by every combination, with no decoder failure.

Decoding the floating-point matrix separates the representation from its storage. The geometric decoder on the signed RP, GASF, GADF and XIRP recover every series to machine precision: the median NMAE never exceeds 1.4e-13 for the signed RP and 5.6e-17, 8.4e-17 and 2.8e-16 for GASF, GADF and XIRP. For these four the whole 8-bit error is therefore due to storage. The only exception is one ESC-50 window that is constant except for two samples, with an amplitude of about 3×10^{-9}.

After 8-bit storage they form a near-lossless group. The signed RP stays between 0.0009 and 0.0022 on six datasets and rises to 0.012–0.017 on Swarm. GASF and GADF are of the same order. XIRP is an order of magnitude lower, because its diagonal stores the series with a fixed range and the decoder averages the proposals of all columns. Per series, the median difference to the signed RP is of order 10^{-3} in favour of GASF, GADF and XIRP on most datasets. The binary and unsigned RP decoders and MTF lose most of the waveform: their NMAE ranges from about 0.03 to 0.34, and the median per-series difference to the signed RP, averaged over datasets, is 0.185 for Hirata on the unsigned RP and 0.311 for MTF.

Table 4: Mean 8-bit NMAE of every encoding–decoder combination at $T = 128$ (500 windows per dataset, every channel). Uns./sig.: unsigned and signed RP.

Dataset	Hirata / uns.	Hirata / sig.	Ours / uns.	Ours / sig.	GASF	GADF	MTF	XIRP
WISDM	0.2411	0.1044	0.1946	0.0010	0.0015	0.0013	0.2838	0.0002
PAMAP2	0.2345	0.1059	0.1899	0.0015	0.0016	0.0013	0.2771	0.0002
ESC-50	0.2177	0.0830	0.1747	0.0018	0.0016	0.0010	0.2725	0.0002
PTB-XL	0.3313	0.1982	0.2874	0.0016	0.0029	0.0023	0.3404	0.0002
Swarm	0.2778	0.0283	0.2547	0.0124	0.0026	0.0015	0.2979	0.0002
STEAD	0.2044	0.0868	0.1771	0.0013	0.0015	0.0011	0.2668	0.0002
EMG	0.1979	0.1205	0.1680	0.0009	0.0013	0.0022	0.2834	0.0006

6.2. Recurrence plots: encoding versus decoder

Table 5 separates the two ingredients of RP reconstruction. On the unsigned RP the geometric decoder recovers the shape of the series exactly but not its reflection: with the reflection chosen by an oracle that knows the original, its error falls to the level of the signed RP. The loss on the unsigned RP is therefore entirely a loss of orientation, which the sign restores without any external information. Hirata's reconstruction does not reach that level on either encoding, because its graph sees only the thresholded recurrence plot. The binary RP itself is the limit: even in floating point and with the oracle orientation, neither decoder brings its error below 0.03 (up to 0.17 across datasets). The sign improves Hirata as well, by choosing the reflection, but on the same signed RP the geometric decoder is one to two orders of magnitude more accurate on six datasets; on Swarm, where the states are nearly collinear, the gap narrows to a factor of about two.

Table 5: Recurrence plots, encoding versus decoder: mean 8-bit NMAE at $T = 128$. "Oracle" picks the reflection using the original series and is an upper bound, not a reconstruction.

	Hirata		Geometric decoder (ours)		
Dataset	Unsigned	Signed	Unsigned	Unsigned, oracle	Signed
WISDM	0.2411	0.1044	0.1946	0.0004	0.0010
PAMAP2	0.2345	0.1059	0.1899	0.0007	0.0015
ESC-50	0.2177	0.0830	0.1747	0.0006	0.0018
PTB-XL	0.3313	0.1982	0.2874	0.0008	0.0016
Swarm	0.2778	0.0283	0.2547	0.0114	0.0124
STEAD	0.2044	0.0868	0.1771	0.0006	0.0013
EMG	0.1979	0.1205	0.1680	0.0003	0.0009

6.3. When is the signed RP decoder exact?

Remark 1 predicts that storage error is amplified when the states (x_i, x_{i+1}) are nearly collinear. Measured by the ratio σ_2/σ_1 of the singular values of the centred states, this is borne out across all datasets. None of the 47 949 series with $\sigma_2/\sigma_1 \geq 0.1$ exceeds an 8-bit NMAE of 0.01, whereas 85% of the series with $\sigma_2/\sigma_1 < 0.01$ do. The share of state pairs whose sign carries direction, by contrast, does not explain the error: slow series have almost all displacements along the diagonal, so their sign is informative in nearly every pair, yet their states are the most collinear. Swarm, sampled at 1 Hz, is the limiting case, and it is the only dataset where the signed RP is not in the near-lossless group.

6.4. Robustness and computational cost

Figure 2a shows how the near-lossless group degrades when the stored image is perturbed. With noise of $\sigma = 16$ intensity levels, the mean 8-bit

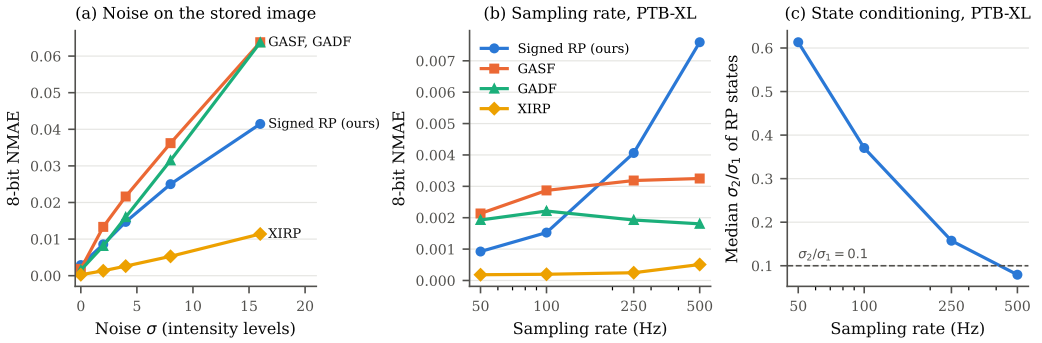


Figure 2: (a) Mean 8-bit NMAE against Gaussian noise added to the stored image ($T = 128$, mean over the seven datasets, two replicates per level). (b) 8-bit NMAE and (c) median conditioning σ_2/σ_1 of the RP states against the sampling rate (PTB-XL, $T = 128$, anti-aliased resampling from 500 Hz). Binary and unsigned RP decoders and MTF stay between 0.08 and 0.34 throughout and are omitted.

NMAE at $T = 128$ rises to 0.011 for XIRP, 0.041 for the signed RP, and 0.064 and 0.064 for GASF and GADF. The signed RP degrades less than both angular fields on 6 of the 7 datasets; on Swarm, whose states are nearly collinear, the three are level. XIRP is the most stable because each sample is read from the diagonal and averaged over the proposals of every column. The lossy combinations are insensitive to the noise, which moves their error by at most 0.019, because their error is already dominated by what the representation discards.

Table 6 reports single-thread cost against window length. Hirata's reconstruction grows fastest, with an exponent of about 2.6 due to all-pairs shortest paths. The geometric decoder on the signed RP is 2.9 times faster than Hirata on the same encoding at $T = 256$ and 5.2 times faster at $T = 1024$. Its measured exponent is below the cubic cost of a dense eigendecomposition, because at these sizes building the matrix and the linear-algebra library dom-

Table 6: Median single-thread time (ms) to encode, store and decode one channel (WISDM, 20 series per length), log–log slope of time against T, and peak traced memory at $T = 1024$.

Decoder	$T = 64$	$T = 256$	$T = 1024$	Exponent	MB at 1024
Hirata / uns.	2.0	58.5	2871.2	2.64	87
Hirata / sig.	2.3	71.8	2880.1	2.59	87
Ours / uns.	0.6	13.1	398.2	2.35	72
Ours / sig.	1.3	25.1	552.0	2.22	72
GASF	0.1	1.0	12.0	1.85	32
GADF	0.1	1.0	13.0	1.72	32
MTF	0.5	1.5	12.8	1.22	24
XIRP	0.1	0.6	12.8	2.03	32

inate. GASF, GADF, XIRP and MTF are one to two orders of magnitude cheaper. Memory grows quadratically for all of them, and cost grows linearly with the number of channels.

6.5. Preservation of classification utility

Table 7 summarises the train-on-original experiment: six classifiers, six labelled datasets and three lengths, 84 combinations on 90 937 windows. ESC-50 is excluded from these combinations because 128 samples at 8 kHz span only 16 ms of sound, which leaves every classifier near chance. Audio is evaluated instead on ESC-10 at window lengths that span 64 to 512 ms (18 000 windows, 36 combinations), with accuracy scored per clip.

The near-lossless group of the fidelity analysis is also the group that preserves utility. The signed RP with the geometric decoder, GASF, GADF and XIRP never lose more than 0.01 of accuracy (or macro-F1 on the multilabel PTB-XL), and their difference to the original is statistically equivalent within

Table 7: Drop from the original accuracy (macro-F1 on PTB-XL) for classifiers trained on original windows and tested on reconstructions. Equivalent: 95% group-bootstrap interval within ±0.01, counted over the single-label combinations.

	Six datasets, $T \in \{64, 128, 256\}$				ESC-10, per clip	
Decoder	Mean drop	Max drop	$\lvert\Delta\rvert \leq 0.01$	Equivalent	Mean drop	Equivalent
Hirata / uns.	0.120	0.376	19%	11/66	0.071	0/36
Hirata / sig.	0.080	0.283	26%	12/66	0.060	0/36
Ours / uns.	0.099	0.359	21%	13/66	0.024	0/36
Ours / sig.	0.000	0.007	100%	63/66	0.001	33/36
GASF	-0.000	0.005	100%	63/66	0.001	34/36
GADF	-0.000	0.002	100%	63/66	0.000	32/36
MTF	0.099	0.400	19%	7/66	0.106	0/36
XIRP	0.000	0.002	100%	65/66	-0.000	36/36

±0.01 in almost every single-label combination. The binary and unsigned RP decoders and MTF lose between 0.08 and 0.12 on average and up to 0.40, and are equivalent in only a minority of combinations. The pattern holds per clip on ESC-10.

6.6. Sampling rate

The sampling rate changes how far apart consecutive samples are, and therefore how collinear the states (x_i, x_{i+1}) become. Figure 2b–c shows the consequence on PTB-XL. From 50 to 500 Hz, the median σ_2/σ_1 falls from 0.61 to 0.08, and the share of series below 0.1 grows from 0% to 67%. The 8-bit NMAE of the signed RP rises accordingly, from 0.0009 to 0.0076, while GASF and GADF stay between 0.0018 and 0.0032 and XIRP is unaffected. The signed RP is thus the most accurate of the three at low rates and the

least accurate at high rates, as Remark 1 predicts. The resampled 100 Hz series and the 100 Hz files distributed with the dataset give the same error (0.0015 and 0.0015), so the effect is not an artefact of the resampling.

7. Discussion

7.1. Three classes of representation

The results sort the compared combinations into three classes that agree across domains, window lengths and evaluation lenses. The first contains the signed RP with the geometric decoder, GASF, GADF and XIRP. Each is exact on the floating-point matrix, so its error after storage is the error of storage, and that error is small enough to leave classifiers trained on original signals unaffected. The second is the unsigned RP: it determines the shape of the series but not its reflection, so its reconstruction is exact up to a sign that no image-only decoder can fix. The third contains the binary RP, under either decoder, and MTF, which discard information the series needs: thresholding keeps only neighbourhoods, and quantile assignment keeps only states. No decoder can recover what these encodings do not store; for them the question is not how to invert but what representative to return.

The classes follow from what each image determines (Table 1), not from the domain. The same ordering holds for accelerometers, electrocardiograms, magnetometers, seismograms, electromyograms and audio, and for every classifier. This matters for image-domain generation: a model trained on the third class cannot produce images that map back to the signals it was trained on, whatever its quality.

7.2. What the sign and the decoder each add

Separating the encoding from the decoder shows that RP reconstruction needs both. The sign is what makes the problem well posed: on the unsigned RP, even an exact geometric decoder loses the orientation. The decoder is what exploits the information the matrix holds. Hirata's reconstruction was designed for the binary plot and sees only its neighbourhoods, so applying it to the signed RP gains the orientation but not the geometry. Using the full distances, the geometric decoder makes the signed RP as accurate as the angular fields on six of the seven datasets.

7.3. Conditioning and sampling rate

The signed RP is the only representation in the first class whose accuracy depends on the signal. Its decoder recovers the two-dimensional geometry of the states (x_i, x_{i+1}), so it is limited by how far that cloud is from a line. Slowly varying windows, whether from Swarm at 1 Hz or from an ECG sampled at 500 Hz, have consecutive samples that are nearly equal and states that are nearly collinear, and there storage error is amplified. The ratio σ_2/σ_1 makes this a measurable condition: above 0.1, no series exceeded a normalised error of 0.01. It also gives a practical rule. For signals sampled far above their bandwidth, the angular fields or XIRP are the safer choice, unless the signal is downsampled first. An embedding delay larger than one sample would also decorrelate the states, but we have not tested it.

7.4. XIRP and the cost of reconstruction

XIRP is the most accurate and the most robust representation in the comparison because it does not encode the series indirectly: the diagonal is

the series, and inversion by mean averages T independent proposals of each sample, which reduces quantisation and pixel noise. Its constraint is that the series must be strictly positive, which the shared calibration provides here. The angular fields and XIRP are also one to two orders of magnitude cheaper to decode than either RP decoder. Among the RP decoders, the geometric decoder is faster than Hirata's and scales better, because a single eigendecomposition replaces all-pairs shortest paths.

7.5. Limitations

- **Controlled inputs.** Every image is the deterministic output of a known encoder. Noise on the stored image measures local stability only; structurally invalid or generated images are not evaluated.
- **Calibration and storage.** All decoders work from the image alone on the normalised domain, but physical units need the per-channel calibration $(x_{\max}, x_{\min})$, which a generative setting would have to supply. Only 8-bit storage is evaluated.
- **Channels.** Every channel is an independent image, so representations that couple channels, and any information shared between them, are outside the study.
- **Sampling of data.** Fidelity uses 500 windows and classification up to 6 000 windows per dataset and length. EMG at 256 samples is not reported (gestures last about three seconds), ESC-50 at 64–256 samples is too short for audio classification and is complemented by ESC-10, and the splits of PTB-XL and ESC-50 are grouped by patient and source recording rather than the official folds.

- **Reference implementation.** Hirata's reconstruction is our implementation of the published algorithm; the conversion from states to samples is ours, and the recurrence threshold is not prescribed by the method.

8. Conclusions

We compared how much of a time series survives its image for the most widely used image representations and for XIRP, on seven datasets from five domains. Four representations are exactly invertible and keep what classifiers need: the signed recurrence plot decoded geometrically, the Gramian angular summation and difference fields, and XIRP. The unsigned recurrence plot loses its orientation, and the binary recurrence plot and the Markov transition field lose most of the waveform under any decoder. For recurrence plots both the sign and a decoder that uses the full distances are necessary, and the conditioning of the states decides when storage error stays negligible, which ties the choice of representation to the sampling rate of the signal. These results give image-domain generation a criterion that comes before any model: only images from an invertible representation can be mapped back to the signals they represent. Evaluating generated and structurally imperfect images is the next step.

CRediT authorship contribution statement

Adriano García-Giralda Milena: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Francisco García Moreno:** Conceptualization, Methodology, Supervision, Writing – review &

editing. **José Antonio Jiménez Madrid:** Methodology, Formal analysis, Writing – review & editing. **María Bermúdez Edo:** Supervision, Writing – review & editing.

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Data availability

All seven datasets are public and are cited at their official sources. The code, window and split manifests, configurations and derived results are available at `https://github.com/adrianoggm/TFG-Tratamiento-de-Series-Temporales` [pending: archived release with DOI, #137].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[pending: List every tool used for writing and for code, and what it was used for (#138).] All AI-assisted content was critically reviewed, verified and revised by the authors, who take full responsibility for the accuracy, originality and integrity of the manuscript.

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